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The Editors
Toxicology
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Dear Editors,

We are pleased to submit our manuscript, “Directional transfer of chemical-gene effects from rodents to humans: an independence-guarded meta-analysis of the Comparative Toxicogenomics Database,” for consideration as a Research Article in *Toxicology*.

Animal-to-human extrapolation is the central validity problem in toxicology, and the field is now pivoting toward human-relevant new approach methodologies. Yet a basic quantity has never been measured at scale: when a chemical is curated to increase or decrease a given gene in a rodent, how often does the same directional effect hold in humans? Using the Comparative Toxicogenomics Database, we assembled 1,165,406 curated chemical-gene-endpoint effects across human, mouse and rat and answered this question directly, comparing each rodent directional call with the directly curated human call for the same chemical and gene.

We believe the study is a good fit for the journal’s emphasis on human-relevant, mechanistic toxicology for the following reasons.

- **A measured human comparator, not an extrapolation.** The primary estimand compares rodent effects with directly curated human effects, so human relevance is built into the design rather than assumed.
- **An independence guard against circular evidence.** Because a naive cross-species comparison is inflated when both calls originate from the same publication, our primary analysis is restricted to effects supported by disjoint PubMed identifiers across species. This lowers apparent agreement from about 89% to about 62%, only marginally above the mouse-versus-rat benchmark, and reframes cross-species divergence as a problem of reproducibility and context dependence rather than phylogenetic distance.
- **A pre-registered external anchor.** Using orthology and graded protein sequence conservation from resources independent of the database, we show that sequence conservation is not associated with directional transfer, addressing a long-standing question about the ortholog conjecture.
- **Actionable framing for evidence integration.** Transfer is governed by molecular endpoint and replication depth rather than by chemical class, suggesting that cross-species evidence should be weighted by what was measured and how firmly it was established.

We confirm that this manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. All authors have approved the submission and agree to be accountable for the work. The study uses only openly available, non-identifiable curated data and secondary resources; it did not involve new experiments in humans or animals. As the analysis is a

meta-analysis of curated literature rather than a controlled exposure study, we have stated explicitly in the Materials and Methods that individual exposure doses, durations and chemical identifiers are properties of the underlying source experiments and are not standardized at the level of this synthesis; our estimands are therefore literature-level directional concordance conditional on curation.

The authors declare no competing interests, and the research received no specific external funding. We have provided highlights, a title page, and comprehensive supplementary material, and the manuscript is prepared in L^AT_EX using the `elsarticle` class with author-year (Harvard) referencing, as accepted by the journal.

Thank you for considering our work. We look forward to the reviewers' comments.

Sincerely,

January G. Msemakweli
On behalf of all co-authors